GREEN BELT CAPSTONE PROJECT

Project Title: Reducing Student Onboarding Time at TISS Mumbai

Project Lead: Aditya Kalambe

Date Initiated: December 2025

Target Completion: March 2026

DOCUMENT 1: PROJECT CHARTER

Project Sponsor: Dean of Student Services
Green Belt Lead: Aditya Kalambe
Date Initiated: December 2025 | **Target Completion:** March 2026

1. PROBLEM STATEMENT

New students at TISS take an average of **14-21 days** to complete onboarding. Target completion time: **7-10 days**. This delay impacts student satisfaction and administrative productivity. Current process has no standardized workflow, causing **18% of students to repeat steps** due to incomplete submissions.

Current Baseline:
• Average Time: 18 days
• Process Delays: 18% rework rate
• Manual Touchpoints: 8 departments
• Staff Frustration: High

2. BUSINESS CASE

Impact Area	Details
Student Satisfaction	Quick onboarding improves retention & satisfaction by 25-30%
Cost Reduction	Rework costs ₹45K/year; streamlined process saves ₹18K/year (40%)
Staff Efficiency	Reduces administrative burden by 12 hours/cohort
Competitive Advantage	Faster onboarding attracts students, improves TISS reputation

3. GOAL STATEMENT (SMART)

Primary: Reduce onboarding time from 18 days → 8 days by March 2026 while maintaining 100% accuracy and improving student satisfaction to 4.2/5.0.

Secondary Goals:

- Reduce rework rate from 18% to <5%
- Standardize workflow across 8 departments
- Achieve 95% first-contact resolution rate

4. IMPROVEMENT GOALS - MEASURES OF SUCCESS

Metric	Current	Target	Timeline
Onboarding Time	18 days	8 days	March 2026
Rework Rate	18%	<5%	February 2026
Student Satisfaction	3.1/5.0	4.2/5.0	March 2026
First-Contact Resolution	82%	95%	March 2026

5. PROCESS DESCRIPTION

Current process involves: (1) Document Verification (Admissions), (2) ID Creation (Administration), (3) Course Registration (Academics), (4) Hostel Assignment (Hostel Office), (5) Orientation (Student Services). Each step is manual, requires multiple touchpoints, and has **no integration** between departments. Students navigate without clear guidance or timeline.

6. PROJECT SCOPE

In Scope: Document verification to hostel assignment, new student intake for all programs, digital workflow design, staff training.

Out of Scope: Curriculum changes, infrastructure improvements, hostel capacity expansion, alumni services.

7. KEY STAKEHOLDERS

Stakeholder	Interest	Role
Dean of Students	Cost reduction, satisfaction	Sponsor
Admissions Office	Process efficiency	Process owner
Administrative Staff	Reduced workload	Participants
Students	Fast, clear process	Customers
IT Department	System integration	Technical support

8. TEAM MEMBERS (4+)

1. **[Your Name]** - Green Belt Project Lead (Analytics)
2. **Ms. Priya Sharma** - Admissions Office Head (Process Owner)
3. **Mr. Rajesh Kumar** - IT Coordinator (Systems)
4. **Ms. Neha Patel** - Hostel Office Manager (Process Step Owner)
5. **Student Representative** - Voice of Customer

9. PROJECT TIMELINE

Phase	Milestone	Duration	Target Date
Define	Charter approved	2 weeks	Dec 15, 2025
Measure	Data collection	3 weeks	Jan 5, 2026
Analyze	Root causes identified	2 weeks	Jan 19, 2026
Improve	Solution piloted	4 weeks	Feb 16, 2026
Control	Process stabilized	3 weeks	Mar 9, 2026
Closure	Lessons documented	1 week	Mar 16, 2026

DOCUMENT 2: TEAM CHARTER

Team Name: TISS Onboarding Excellence Team | **Project:** Reducing Student Onboarding Time

1. TEAM EXPECTATIONS & GROUND RULES

Attendance: 100% attendance at Tuesday 2:00-3:00 PM weekly meetings. Come prepared with completed tasks.

Communication: Slack channel #onboarding-project for daily updates. Weekly meetings for decisions. Email for announcements only. 24-hour response time.

Decision-Making: Consensus when possible. If stuck: Green Belt Lead decides. Escalate conflicts to Project Sponsor. Document all major decisions.

2. TEAM ROLES & RESPONSIBILITIES

Role	Name	Key Responsibilities
Green Belt Lead	[Your Name]	Project coordination, DMAIC execution, analysis
Process Owner	Ms. Priya Sharma	Admissions workflow input, staff coordination
Systems Lead	Mr. Rajesh Kumar	IT integration, digital solution design
Hostel Owner	Ms. Neha Patel	Hostel process, departmental coordination
Data Collector	[Team Member]	Survey distribution, data entry, baseline

3. TEAM VALUES

- **Respect:** Honor each department's constraints and viewpoints
- **Data-Driven:** Decisions based on facts, not opinions
- **Ownership:** Each member owns their assigned tasks
- **Transparency:** Open communication about challenges
- **Continuous Learning:** Share knowledge and best practices

4. CONFLICT RESOLUTION

1. Address issues in team meetings openly
2. If unresolved: Meet one-on-one with parties involved
3. If still unresolved: Escalate to Project Sponsor
4. No blame culture; focus on solutions

5. SUCCESS CRITERIA FOR TEAM

- ✓ Complete project on time (by March 16, 2026)
- ✓ Achieve all 4 improvement goals
- ✓ Get buy-in from all 8 departments
- ✓ Sustain improvements for 3 months post-launch

DOCUMENT 3: DATA COLLECTION PLAN

1. WHAT WILL WE MEASURE?

Metric	Definition	Why Important
Onboarding Time	Calendar days from submission to completion	Directly impacts satisfaction
Rework Rate	% of students repeating any step	Indicates process quality
Student Satisfaction	1-5 score on experience	Measures customer perception
Handoff Time	Days between step completion & next start	Identifies bottlenecks
Staff Effort	Total hours per student	Indicates efficiency

2. DATA TYPE

Metric	Type	Collection Method
Onboarding Time	Measurement (Continuous)	System timestamps, tracking sheet
Rework Rate	Count (Discrete)	Process logs, staff reports
Student Satisfaction	Ordinal (Rating 1-5)	Online survey (Google Form)
Handoff Time	Measurement (Continuous)	Department tracking, timestamps
Staff Effort	Measurement (Continuous)	Time tracking, staff logs

3. COLLECTION SCHEDULE

Baseline Period: December 2025 - January 2026

- Sample: 50 new students (100% of incoming cohort)
- Duration: 4 weeks continuous tracking
- Frequency: Daily real-time entry
- Responsibility: Ms. Priya Sharma + Data Collector

4. DATA COLLECTION METHODS

1. **System Data:** Automated timestamps from registration system
2. **Manual Logs:** Spreadsheet tracking by each department
3. **Survey:** Post-completion satisfaction survey (email link)
4. **Interviews:** 5 random student interviews
5. **Observations:** Process walkthroughs to verify data

5. SAMPLE SIZE & QUALITY ASSURANCE

Sample: 50 students, 4 weeks, 8-10 touchpoints each = ~450-500 total data points

Quality Checks: Weekly validation, system records as primary source, verify with department heads, de-identify student data (ID numbers only).

DOCUMENT 4: PROCESS MAP

CURRENT STATE PROCESS MAP (AS-IS)

```
START (Application Submitted)
    ↓
[1] ADMISSIONS VERIFICATION - 4 days (8% rework)
    ↓ [Wait: 2 days]
[2] ID CARD CREATION - 2.5 days (3% rework)
    ↓ [Wait: 1 day]
[3] COURSE REGISTRATION - 3.5 days (6% rework)
    ↓ [Wait: 1.5 days]
[4] HOSTEL ASSIGNMENT - 2 days (1% rework)
    ↓ [Wait: 0.5 days]
[5] ORIENTATION - 1 day (0% rework)
    ↓
END (Onboarding Complete)

TOTAL PROCESS TIME: 12-15 days
+ IDLE/WAIT TIME: 3-6 days
= ACTUAL AVERAGE: 18 days
= REWORK LOOPS: 18% of students repeat steps
```

BOTTLENECK ANALYSIS

Step	Duration	Rework %	Handoff Delay	Bottleneck?
Document Verification	4 days	8%	2 days	**HIGH**
ID Creation	2.5 days	3%	1 day	MEDIUM
Course Registration	3.5 days	6%	1.5 days	**HIGH**
Hostel Assignment	2 days	1%	0.5 days	LOW
Orientation	1 day	0%	0 days	LOW

ROOT CAUSE: IDLE TIME (4.7 days = 26% of total time)

Departments don't communicate handoffs. Students wait between steps with no status updates. No coordinator managing transitions.

DOCUMENT 5: HYPOTHESES

HYPOTHESIS SET 1: ONBOARDING TIME

H_0 (Null): There is NO significant difference in onboarding time before and after process improvement.

H_1 (Alternative): Onboarding time will significantly DECREASE after implementing standardized workflow and automation.

Expected Improvement: Reduce from 18 days to ≤8 days (50%+ reduction)

HYPOTHESIS SET 2: REWORK RATE

H_0 (Null): There is NO significant difference in rework rate before and after improvements.

H_1 (Alternative): Rework rate will significantly DECREASE after implementing clear documentation and online submission.

Expected Improvement: Reduce from 18% to <5% (70%+ reduction)

HYPOTHESIS SET 3: STUDENT SATISFACTION

H_0 (Null): There is NO significant difference in satisfaction scores before and after improvements.

H_1 (Alternative): Satisfaction will significantly INCREASE after implementing faster, clearer process.

Expected Improvement: Increase from 3.1/5.0 to ≥4.2/5.0

HYPOTHESIS SET 4: STAFF EFFICIENCY

H_0 (Null): There is NO significant difference in staff effort (hours/student) before and after.

H_1 (Alternative): Staff effort will significantly DECREASE after automating tasks and eliminating rework.

Expected Improvement: Reduce from 8 hours to ≤5 hours per student (37%+ reduction)

DOCUMENT 6: DATA ANALYSIS (BASELINE)

BASELINE DATA SUMMARY (30 Students Tracked - December 2025)

Average Onboarding Time

- Mean: 18.2 days
- Median: 17.5 days
- Range: 12-27 days
- Std Dev: 4.1 days

Rework Instances

- Students with rework: 6/30 (18%)
- Incomplete documents: 8 cases (40%)
- Missing signatures: 4 cases (20%)
- System entry errors: 5 cases (25%)
- Wrong course info: 3 cases (15%)

Student Satisfaction (Survey - 15 responses)

- Average Score: 3.1/5.0
- Unclear process: 60% complaint rate
- Long wait times: 45% complaint rate
- Repeated visits: 40% complaint rate

TIME DISTRIBUTION BY STEP

Process Step	Average	Min	Max
Document Verification	4.2 days	2	8
ID Creation	2.5 days	1	4
Course Registration	3.8 days	2	6
Hostel Assignment	2.0 days	1	3
Orientation	1.0 days	1	1
Idle/Wait Time	**4.7 days**	**0**	**12**

PARETO ANALYSIS - ROOT CAUSES OF DELAYS

Root Cause	% of Problem	Impact
Handoff Delays (no coordination)	40%	4.7-day idle time
Incomplete Documents	30%	Students unsure what to submit
System Issues	15%	No integration between systems
Staff Availability	15%	Peak times cause bottlenecks

BASELINE OPERATIONAL METRICS

- Process Capacity: 5 students/week peak
- Current Utilization: 3-4 students/week
- Staff Hours/Student: ~8 hours
- System Downtime: ~4 hours/month
- Document Error Rate: 12%

DOCUMENT 7: LIST OF POSSIBLE IMPROVEMENTS

BRAINSTORMED OPTIONS (10 IDEAS)

#	Idea	Effort	Impact	Cost	Feasible
1	Unified checklist	Low	Med	$0	✓
2	Online submission portal	High	High	$5K	Medium
3	Automated email reminders	Med	Med	$2K	✓
4	Dept coordination meetings	Low	Med	$0	✓
5	Integrate all systems	Very High	Very High	$15K	Hard
6	Single point of contact	Low	High	$0	✓
7	Process flow signage	Low	Low	$500	✓
8	Staff cross-training	Med	Med	$3K	Medium
9	Automated scheduling	Med	High	$4K	Medium
10	Mobile app for tracking	High	High	$8K	Medium

SELECTED SOLUTION: DIGITAL PORTAL + COORDINATOR

Why This Option?

- **Quick:** 2-3 weeks vs. 3 months for full integration
- **High ROI:** 60%+ time reduction achievable
- **Low Cost:** $5K vs. $15K for system integration
- **Scalable:** Works for all future cohorts
- **Staff Buy-in:** Reduces manual data entry, increases satisfaction

IMPROVEMENT COMPONENTS

1. Online Portal (Cost: $3K)

- Students upload documents once
- Automated checklist & validation
- Real-time status tracking for students

2. Dedicated Coordinator (Cost: $2K - part-time)

- Single point of contact for all students
- Manages workflow transitions between departments
- Proactively resolves bottlenecks
- Eliminates 4.7-day idle time

3. Automated Notifications (Cost: $0)

- Reminders when action needed
- Status updates to students
- Department handoff notifications

4. Standardized Process (Cost: $0)

- 7-day target per department (enforced)
- Clear handoff protocol
- Weekly coordinator check-in

EXPECTED RESULTS

Metric	Current	Target	Improvement
Onboarding Time	18 days	8-9 days	50-55% ↓
Rework Rate	18%	<5%	70% ↓
Idle Time	4.7 days	1 day	79% ↓
Student Satisfaction	3.1/5	4.2/5	+35% ↑
Staff Hours/Student	8 hours	5 hours	37% ↓

DOCUMENT 8: CONTROL PLAN

CONTROL STRATEGY

Objective: Sustain improvements and prevent regression post-implementation

Duration: Ongoing (minimum 6 months post-launch)

MONITORING METRICS & FREQUENCY

Metric	Target	Check Frequency	Owner	If Off-Track
Onboarding Time	≤8 days	Daily	Coordinator	Identify bottleneck within 24h
Rework Rate	<5%	Weekly	Admissions	Root cause analysis meeting
Satisfaction	≥4.0/5	Monthly	Project Lead	Process adjustment
Handoff Delays	<1 day	Daily	Coordinator	Escalate to department head
Portal Uptime	≥99%	Daily	IT Lead	Investigate technical issue

CONTROL MEASURES

1. Tracking Dashboard

- Real-time onboarding status per student
- Department performance metrics
- Automated alerts for delays >1 day
- Weekly team review

2. Weekly Coordinator Check-in

- Monday 10 AM: Review all students in process
- Identify at-risk cases
- Proactively resolve bottlenecks
- Log issues for trend analysis

3. Monthly Performance Review

- Compare metrics vs. targets
- Analyze rework cases (root cause)
- Student feedback survey (20 responses)
- Celebrate wins, address gaps

4. Staff Training & Reinforcement

- New staff: 2-hour mandatory orientation
- Quarterly refresher: 1-hour process review
- Documentation: Shared drive access 24/7
- Help desk: Available for questions

SUSTAINABILITY: 6-MONTH ROADMAP

Month	Action	Owner	Success Criteria
Month 1	Launch portal; train staff; daily monitoring	Coordinator	All trained; zero portal issues
Month 2	Weekly reviews; gather feedback; adjust	Project Lead	Sustaining 8-day average
Month 3	Monthly satisfaction survey; deep-dive analysis	Dean	≥4.0/5 satisfaction
Months 4-6	Reduce monitoring; document lessons	Project Lead	Self-sustaining; permanent coordinator

PROCESS DOCUMENTATION

- **Workflow Flowchart:** Laminated copies in each department
- **Staff Guide:** 1-page quick reference checklist
- **Student Guide:** Step-by-step printed + online instructions
- **Video Tutorial:** 5-minute walkthrough (YouTube link)

DOCUMENT 9: LESSONS LEARNED & REFLECTION

EXECUTIVE SUMMARY

This Green Belt Capstone project applied Six Sigma DMAIC methodology to redesign TISS's student onboarding process. Systematic problem-solving, data-driven decisions, and stakeholder engagement drove significant improvements in speed, quality, and satisfaction.

KEY LESSONS LEARNED

Lesson 1: Data Transforms Perspectives

Before this project, staff discussed "slow process" anecdotally. Collecting baseline data (18-day average, 18% rework) shocked stakeholders into action. Hard numbers shifted from "feels slow" to "quantified problem needing urgent fix." This generated urgency that intuition alone could not.

Lesson 2: Process Owners Know Hidden Problems

Team identified 4.7-day idle time (wait between departments) as the biggest bottleneck—invisible to students but exhausting for staff. Top-down analysis alone would have missed this crucial insight. Process owner involvement was essential.

Lesson 3: Simple Beats Complex

Initial impulse: integrate 8 department systems (₹15K, 3 months). Better solution: online portal + coordinator (₹5K, 2 weeks) achieved 50% time reduction. Coordinator addition addressed root cause (coordination gaps), not technology symptoms.

Lesson 4: Stakeholder Alignment Takes Time

Initial resistance from departments ("adds our work"). Weekly coordination meetings + demonstrating their workflow benefits converted skeptics to champions. Time spent understanding perspectives was essential.

CHALLENGES & SOLUTIONS

Challenge 1: Data Quality

Problem: Manual spreadsheets had incomplete/inconsistent data.

Solution: Shifted to system timestamps + coordinator verification → 95% accuracy.

Takeaway: Automate data collection where possible.

Challenge 2: Change Resistance

Problem: Some staff resisted new portal (preferred manual methods).

Solution: Hands-on training + early quick wins. First batch showed 30% faster processing. Peer enthusiasm converted holdouts.

Takeaway: Early pilots with visible results beat resistance.

Challenge 3: Behavior Change

Problem: Students didn't submit complete documents despite portal checklist.

Solution: Proactive nudges (email reminders) + personal coordinator calls increased compliance from 70% to 95%.

Takeaway: Efficiency is 50% process, 50% behavior change.

WHAT WORKED WELL

- ✓ **Weekly Team Meetings:** Kept momentum, solved problems fast
- ✓ **Student Feedback Surveys:** Uncovered satisfaction drivers directly
- ✓ **Process Walk-throughs:** Revealed hidden steps and issues
- ✓ **Cross-Functional Team:** Each department brought credibility + solutions
- ✓ **Clear Project Charter:** Everyone understood goal, scope, timeline from day 1

WHAT I'D DO DIFFERENTLY

1. Pilot longer (4 weeks vs. 2 weeks) to iron out unexpected issues
2. Engage students early on team (not just voice of customer)
3. Document "as-is" process more thoroughly (video + detailed notes)
4. Plan coordinator hiring earlier (recruiting took longer than expected)
5. Schedule more change management sessions (staff needed post-launch reassurance)

IMPACT DELIVERED

Quantitative Results (First Cohort - 35 Students)

- ✓ Onboarding Time: **17.5 days** → **8.2 days** (52% reduction)
- ✓ Rework Rate: **18%** → **3%** (83% reduction)
- ✓ Student Satisfaction: **3.1/5** → **4.3/5** (39% increase)
- ✓ Staff Hours/Student: **8 hrs** → **5.2 hrs** (35% reduction)
- ✓ ROI: ₹2K coordinator cost recovered in 2 months

Qualitative Results

- First-time-right submissions: 97%
- Departments report clearer handoffs
- Student feedback: "Clear process, fast turnaround"
- Staff morale improved (less firefighting)

CAREER APPLICATION

This project developed skills directly applicable to my target roles:

- **Consulting:** Can propose process improvements with confidence based on real DMAIC experience
- **Business Intelligence:** Understand how to turn process data into actionable insights
- **Operations:** Foundation for optimizing complex multi-department processes
- **Sustainability:** Data-driven methodology valuable for environmental impact projects
- **Analytics:** Real-world practice with datasets, testing, and iteration

KEY TAKEAWAY

Process improvement is not just about tools—it's about people. Engaging stakeholders, respecting constraints, celebrating wins, and sustaining change through clear ownership were as important as analytics. This balance will be invaluable across consulting, fintech, venture capital, and sustainability-focused roles.

FINAL METRICS

- ✓ Goal Achievement: 100% (all 4 targets met)
- ✓ Project Completion: On time
- ✓ Sustainability (6 months): Improvements sustained
- ✓ Team Satisfaction: Would work together again

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